

Multiple Choice (10 marks)

1. The _____ is anything which your search engine cannot search.
 - (a) Haunted web
 - (b) World Wide Web
 - (c) Surface web
 - (d) Deep Web
2. A hash function guarantees the integrity of a message. It guarantees that the message has not be
 - (a) Replaced
 - (b) Overview
 - (c) Changed
 - (d) Violated
3. When the data must arrive at the receiver exactly as they were sent, its called
 - (a) Message Condentiality
 - (b) Message Integrity
 - (c) Message Splashing
 - (d) Message Sending
4. A packet filter firewall filters at the
 - (a) Application or transport
 - (b) Data link layer
 - (c) Physical Layer
 - (d) Network or transport layer
5. The Message condentiality or privacy means that the sender and the receiver expect
 - (a) Integrity
 - (b) Condentiality
 - (c) Authentication
 - (d) Nonrepudiation

True / False (10 marks)

Answer True or False

6. True or False: A buffer overflow occurs when the program detects a filled buffer and begins to reject additional requests for memory.
7. True or False: WEP is considered a reliable wireless security protocol due to its encryption capabilities.
8. True or False: The mishandling of undefined variables is a common vulnerability within the transport layer protocols.
9. True or False: A buffer-overflow attack exploits vulnerabilities by overflowing a buffer to inject malicious instructions into memory.
10. True or False: The .Net framework is primarily designed for building secure applications and does not facilitate vulnerability exploitation.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss why a compiler cannot determine the absolute address of a local variable at compile time, particularly in relation to stack allocation and function call dynamics.
12. Discuss the classic security properties in cybersecurity, identifying which one is not traditionally included and analyzing its significance in the context of overall security frameworks.

End of Paper